

Lesson 5 · Mixed Practice: Exam Guide

NCEA Level 2 Mathematics · AS 91262 Apply calculus methods in solving problems · Handout

1 The one idea

Every question is one of a few types. **Read the wording, name the method, then write the calculus.**
The words “use calculus” mean the derivative or anti-derivative must appear in your working.

2 Choosing the method

the question says	method
gradient at $x = \dots$	differentiate, substitute x into $f'(x)$
point where the gradient is ...	solve $f'(x) = m$; y from $f(x)$
equation of the tangent	point from f , gradient from f' , $y - y_1 = m(x - x_1)$
turning point, maximum, minimum	$f'(x) = 0$; y from f ; justify with f''
increasing or decreasing	$f'(x) > 0$ or $f'(x) < 0$, an interval
largest / smallest (words)	build a formula in one variable, then optimise
sketch the gradient function	zeros under turning points; sign from up or down
find $f(x)$ given $f'(x)$ and a point	anti-differentiate, $+c$, use the point
rate of change	derivative at the time; meaning and units
velocity, acceleration from s	differentiate once, twice
displacement from v or a	anti-differentiate, one constant each time
maximum height, at rest	$v = 0$

3 All the rules

differentiate	anti-differentiate
$\frac{d}{dx}(ax^n) = anx^{n-1}$	$ax^n \rightarrow \frac{ax^{n+1}}{n+1} + c$
constant $\rightarrow 0$; $kx \rightarrow k$	$k \rightarrow kx + c$
expand brackets first	find c from a point or a starting value
$f'(x)$: differentiate twice	$s \leftarrow v \leftarrow a$
$s \rightarrow v \rightarrow a$	

4 Answer frames for Merit and Excellence

Justify a maximum. “ $\frac{d^2A}{dx^2} = -4 < 0$, so the area is a maximum when $x = 15$.”

Interpret a rate. “At $t = 6$ the number of spectators is decreasing at 528 people per hour.”

Two unknowns. “On the curve: ... (equation 1). Gradient: ... (equation 2). Solving: ...”

Building a model (Excellence)

1. Name the variables: “Let x be ..., h be ...”
2. Write the constraint as an equation and rearrange for one variable.
3. Write the quantity to optimise in one variable.
4. Differentiate, set $= 0$, solve; reject impossible values with a reason.
5. Justify maximum or minimum, then answer the question in a sentence with units.

Worked example · open box (Excellence)

An open box has a square base of side x cm and height h cm. Its outside surface area is 108 cm^2 . Find the maximum volume.

Step 1 Surface: $x^2 + 4xh = 108$, so $h = \frac{108 - x^2}{4x}$.

Step 2 Volume: $V = x^2h = \frac{x(108 - x^2)}{4} = 27x - \frac{1}{4}x^3$.

Step 3 $\frac{dV}{dx} = 27 - \frac{3}{4}x^2 = 0$, so $x^2 = 36$, $x = 6$ (lengths are positive).

Step 4 $\frac{d^2V}{dx^2} = -\frac{3}{2}x = -9 < 0$, so maximum. Then $h = \frac{108 - 36}{24} = 3$.

Step 5 Maximum volume $6 \times 6 \times 3 = 108 \text{ cm}^3$.

5 What each grade looks like

From the standard's criteria: Achievement is *apply calculus methods in solving problems*; Merit adds *relational thinking*; Excellence adds *extended abstract thinking*.

Achieved	one correct method with working: a gradient, a tangent, a turning point, $f(x)$ with c , a rate.
Merit	linked steps: unknown constants, justified max or min, an interval, a rate found from a value, $a \rightarrow v \rightarrow s$.
Excellence	a model built from words, optimised and justified; or a general result with letters; with clear communication.

6 Exam habits

- Attempt every part: an earlier answer can still earn credit even if a later step goes wrong.
- Write the derivative on its own line before substituting.
- Keep 3 or 4 significant figures until the last step; round the final answer and give units.
- Check with the fx-9750GIII (GRAPH $\rightarrow$ G-SOLVE, d/dx in RUN-MATRIX), but never instead of the working.

7 Ways to lose marks

From the material plus teaching observation, across the whole standard.

- L1** No calculus shown: write $f'(x)$ (or v , $\frac{dP}{dt}$...) before substituting.
- L2** Wrong function: gradients and rates from the derivative, values and heights from the original.
- L3** Incomplete answer: a point needs x and y ; an optimisation answer needs the quantity asked.
- L4** "Maximum" or "minimum" with no test: use f'' or the gradient either side, in a sentence.
- L5** $+c$ forgotten, or not found from the given point or starting value.
- L6** Rates and motion without meaning and units ("decreasing at 528 people per hour").